BIRZEIT UNIVERSITY

Computer Vision - MSAI6330

Assignments #2

Submission deadline: 27.11.2026

Objective

The objective of this assignment is to implement image matching using local invariant features and use the resulting correspondences to estimate a homography and create an image panorama.

Submission Requirements

- Write Python code to complete all the tasks.
- Use a **Jupyter Notebook (.ipynb)** to implement and document your solutions.
- Include the input images, processed results, and appropriate visualizations.
- Add brief explanations of the methods used and observations about the results.
- Submit the completed Jupyter Notebook.

Tasks:

1. **Feature Detection:** Load the attached two images and detect feature points in both images using a feature detector such as Harris corners or blob detection.

2. **Feature Description:** Extract descriptors for the detected keypoints. You need to test two types of descriptors:
   a. SIFT**:** Use the SIFT descriptor.
   b. Intensity-Based Descriptor: Extract a local image patch (e.g. 11x11) around each keypoint and flatten its pixel values into a one-dimensional vector.

3. **Descriptor Matching:** Compute the pairwise distances between descriptors from the two images using Euclidean distance.

4. **Putative Match Selection:** Select candidate matches based on descriptor distances using a pre-defined threshold or by selecting the top matches with the smallest distances. Explain the selected threshold or matching strategy.

5. **Match Visualization:** Display the selected matches by placing the two images side by side and drawing lines connecting corresponding keypoints.

6. **Homography Estimation Using RANSAC:** Implement RANSAC to estimate the homography between the two images. Report:
    - The number of inlier matches.
    - The average squared reprojection error of the inliers. The error is calculated by transforming keypoints from one image into the coordinate system of the other image and measuring the squared Euclidean distance between corresponding points.
    - A visualization showing the inlier correspondences in both images.

7. **Panorama Creation:** Create a new image big enough to hold the panorama and composite the two images into it. You can composite by averaging the pixel values where the two images overlap, or by using the pixel values from one of the images.

8. Compare the performance of SIFT and intensity-based descriptors in terms of the number of putative matches, number of RANSAC inliers, reprojection error, and final panorama quality.

General Guidelines

- You should use Harris corner or blob detection to obtain keypoints for the intensity-based descriptor. For the SIFT experiment, you may use SIFT to detect keypoints and extract descriptors.
- You can use the OpenCV library to extract keypoints and compute descriptors through the function cv2.SIFT_create().detectAndCompute.
- You can use the functions getPerspectiveTransform to calculate a homography, and warpPerspective to apply it on an image.
- Ensure that the notebook cells are organized, executable, and accompanied by brief explanations. Include a short conclusion summarizing the main findings.

Rubrics

Component	**Marks**
Feature Detection	10
Feature Description	15
Descriptor Matching	15
Match Visualization	10
Homography Estimation (RANSAC)	20
Panorama Creation	15
Notebook organization, explanations, and conclusion	15
Total	100